

# Column Space and Nullspace

Two questions every matrix answers:  
**which targets can I reach? and which  
trades change nothing?**

With portfolios, hedges and put–call parity.

$C(A)$  and  $N(A)$

$$x = x_p + x_n$$

MIT 18.06, Lecture 6

# Funds as columns

Rows: budget (₹ lakh), return (₹ thousand), exposure. Columns: Nifty, bonds, and a **balanced** fund (half Nifty, half bonds).

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$$

|          | Column space $C(A)$        | Nullspace $N(A)$             |
|----------|----------------------------|------------------------------|
| Question | Which targets can I reach? | Which trades change nothing? |
| Made of  | targets $b$                | trades $x$                   |

# **$Ax = b$ is solvable exactly when $b \in C(A)$**

$Ax$  is always a mix of the columns:

$$A \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 9.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix}$$

So a target can be hit only if it's a mix of the columns. Balanced =  $\frac{1}{2}$  Nifty +  $\frac{1}{2}$  bonds adds no new direction:  $C(A)$  stays the plane  $R = 7B + 5E$ .

# Twin index funds

Two Nifty funds from different houses: both columns  $(1, 12)$ .

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 - 1 \\ 12 - 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Move ₹1 lakh from one twin to the other: **nothing changes**. All multiples  $c(1, -1)$  form the nullspace, a line.

# Find it by elimination

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

No pivot in column 3, so balanced is **free**. Choose  $z = -2$ : then  $y = 1$ ,  $x = 1$ .

Check:  $1 + 1 - 2 = 0$ ,  $12 + 7 - 19 = 0$ ,  $1 + 0 - 1 = 0$  ✓

## MEANING

Buy ₹1L Nifty + ₹1L bonds, sell ₹2L balanced: you built the balanced fund yourself.

# Do-nothing + do-nothing = do-nothing

If  $Av = 0$  and  $Aw = 0$ :

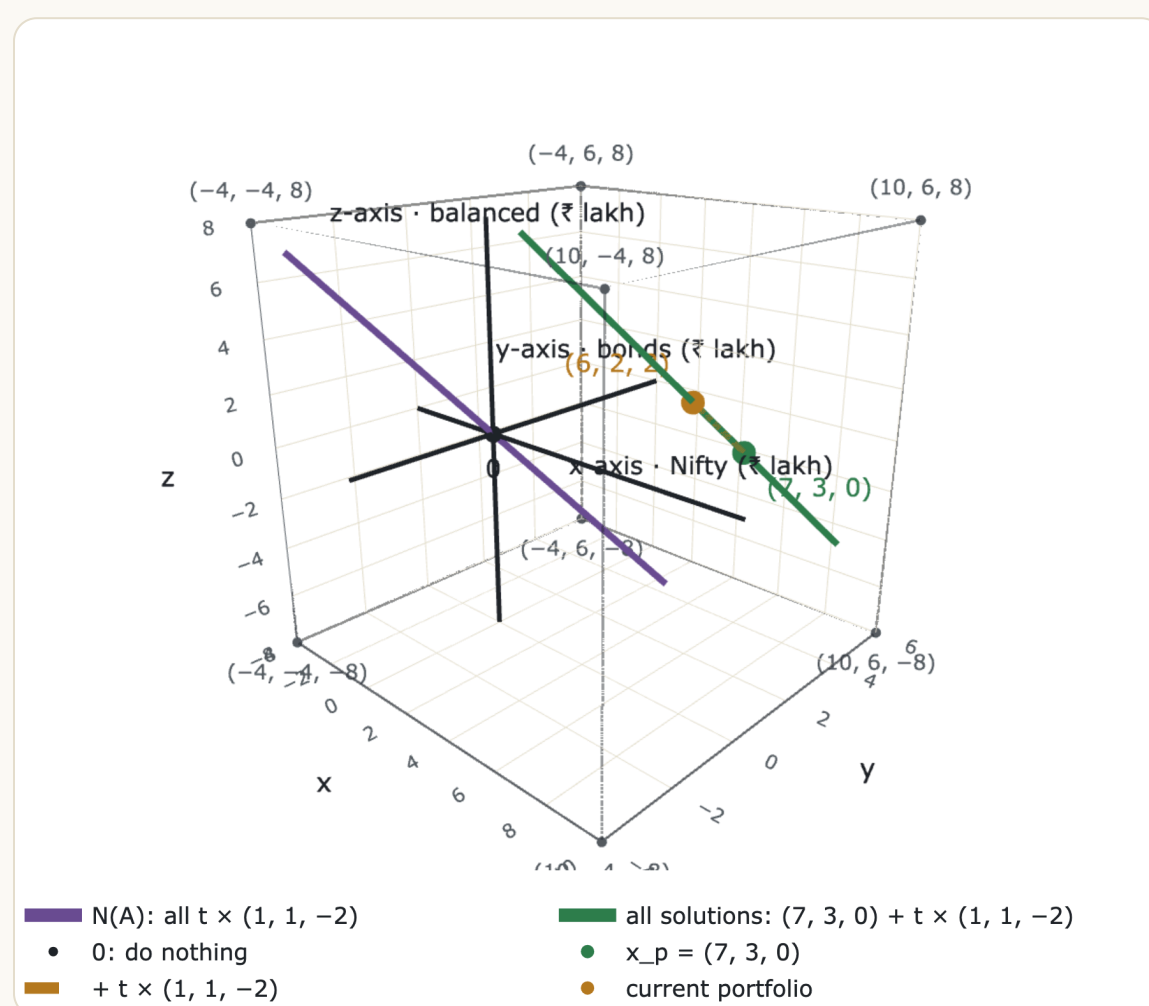
$$A(v + w) = 0 + 0 = 0 \quad A(cv) = c \cdot 0 = 0$$

So  $N(A)$  is always a subspace, of the **trade** space  $\mathbb{R}^n$ .  $C(A)$  is a subspace of the **target** space  $\mathbb{R}^m$ .

Swap balanced for midcap and elimination finds 3 pivots:  $N(A) = \{0\}$ , and every target has **exactly one** portfolio.

# One answer + the nullspace

$$x = \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$



$(7, 3, 0)$ ,  $(6, 2, 2)$ ,  $(4, 0, 6)$  all give ₹10L, ₹105K, exposure 7.



# The solutions are a shifted line

$(0, 0, 0)$  isn't a solution: doing nothing has budget 0, not 10.

Add two solutions:  $(7, 3, 0) + (6, 2, 2) = (13, 5, 2)$   
hits  $(20, 210, 14)$ , which is  $2b$ , not  $b$ .

## RULE

Linear rules " $= 0$ " give a subspace (a nullspace). A non-zero right side shifts it off the origin.



# Put-call parity is a nullspace

Payoff per lot at 19,800 / 20,000 / 20,200: futures  
(−200, 0, 200), 20,000 call (0, 0, 200), 20,000 put  
(200, 0, 0).

$$1 \cdot \text{fut} - 1 \cdot \text{call} + 1 \cdot \text{put} = (0, 0, 0)$$

So "1 future + 2 calls" and "3 futures + 2 puts" pay the same everywhere. If one costs more, sell it and buy the other.

# Key takeaways

1.  $Ax = b$  solvable  $\Leftrightarrow b \in C(A)$ .
2.  $N(A)$  = all  $x$  with  $Ax = 0$ : do-nothing trades.
3. Free columns in elimination  $\Rightarrow$  non-zero nullspace.
4.  $N(A) = \{0\} \Rightarrow$  one portfolio per target.
5. All solutions =  $x_p$  + nullspace.
6. Solutions of  $Ax = b \neq 0$  are not a subspace.

QUANT FROM FIRST PRINCIPLES

# The full interactive lesson is free

A nullspace lab for your own funds and targets, a slider through every solution, and worked hedging and options examples:

**[learn.krisarj.com](https://learn.krisarj.com)**

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